

Packet Tracer - Connect a Wired and Wireless LAN

Addressing Table

Device	Interface	IP Address	Connects To
Cloud	Eth6	N/A	F0/0
	Coax7	N/A	Port0
Cable Modem	Port0	N/A	Coax7
	Port1	N/A	Internet
Router0	Console	N/A	RS232
	F0/0	192.168.2.1/24	Eth6
	F0/1	10.0.0.1/24	F0
	Ser0/0/0	172.31.0.1/24	Ser0/0
Router1	Ser0/0	172.31.0.2/24	Ser0/0/0
	F1/0	172.16.0.1/24	F0/1
WirelessRouter	Internet	192.168.2.2/24	Port 1
	Eth1	192.168.1.1	F0
Family PC	F0	192.168.1.102	Eth1
Switch	F0/1	172.16.0.2	F1/0
Netacad.pka	F0	10.0.0.254	F0/1
Configuration Terminal	RS232	N/A	Console

Objectives

Part 1: Connect to the Cloud

Part 2: Connect Router0

Part 3: Connect Remaining Devices

Part 4: Verify Connections

Part 5: Examine the Physical Topology

Background

When working in Packet Tracer (a lab environment or a corporate setting), you should know how to select the appropriate cable and how to properly connect devices. This activity will examine device configurations in Packet Tracer, selecting the proper cable based on the configuration, and connecting the devices. This activity will also explore the physical view of the network in Packet Tracer.

Instructions

Part 1: Connect to the Cloud

Step 1: Connect the cloud to Router0.

- a. At the bottom left, click the orange lightning icon to open the available **Connections**.
- b. Choose the correct cable to connect **Router0 F0/0** to **Cloud Eth6**. **Cloud** is a type of switch, so use a **Copper Straight-Through** connection. If you attached the correct cable, the link lights on the cable turn green.

Step 2: Connect the cloud to Cable Modem.

Choose the correct cable to connect **Cloud Coax7** to **Modem Port0**.

If you attached the correct cable, the link lights on the cable turn green.

Part 2: Connect Router0

Step 1: Connect Router0 to Router1.

Choose the correct cable to connect **Router0 Ser0/0/0** to **Router1 Ser0/0**. Use one of the available **Serial** cables.

If you attached the correct cable, the link lights on the cable turn green.

Step 2: Connect Router0 to netacad.pka.

Choose the correct cable to connect **Router0 F0/1** to **netacad.pka F0**. Routers and computers traditionally use the same wires to transmit (1 and 2) and receive (3 and 6). The correct cable to choose consists of these crossed wires. Although many NICs can now autosense which pair is used to transmit and receive, **Router0** and **netacad.pka** do not have autosensing NICs.

If you attached the correct cable, the link lights on the cable turn green.

Step 3: Connect Router0 to the Configuration Terminal.

Choose the correct cable to connect **Router0 Console** to **Configuration Terminal RS232**. This cable does not provide network access to **Configuration Terminal**, but allows you to configure **Router0** through its terminal.

If you attached the correct cable, the link lights on the cable turn black.

Part 3: Connect Remaining Devices

Step 1: Connect Router1 to Switch.

Choose the correct cable to connect **Router1 F1/0** to **Switch F0/1**.

If you attached the correct cable, the link lights on the cable turn green. Allow a few seconds for the light to transition from amber to green.

Step 2: Connect Cable Modem to Wireless Router.

Choose the correct cable to connect **Cable Modem Port1** to **Wireless Router Internet** port.

If you attached the correct cable, the link lights on the cable will turn green.

Step 3: Connect Wireless Router to Family PC.

Choose the correct cable to connect **Wireless Router Ethernet 1** to **Family PC**.

If you attached the correct cable, the link lights on the cable turn green.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping netacad.pka

Pinging 10.0.0.254 with 32 bytes of data:
Reply from 10.0.0.254: bytes=32 time=2ms TTL=126
Reply from 10.0.0.254: bytes=32 time=2ms TTL=126
Reply from 10.0.0.254: bytes=32 time=2ms TTL=126
Reply from 10.0.0.254: bytes=32 time=1ms TTL=126

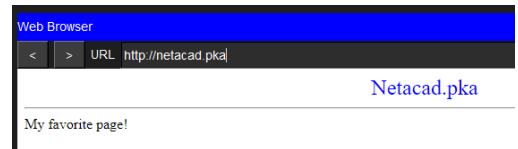
Ping statistics for 10.0.0.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```

Part 4: Verify Connections

Step 1: Test the connection from Family PC to netacad.pka.

- Open the **Family PC** command prompt and ping **netacad.pka**.
- Open the **Web Browser** and the web address **http://netacad.pka**.



Step 2: Ping the Switch from Home PC.

Open the **Home PC** command prompt and ping the **Switch** IP address of to verify the connection.

Step 3: Open Router0 from Configuration Terminal.

- Open the **Terminal** of **Configuration Terminal** and accept the default settings.
- Press **Enter** to view the **Router0** command prompt.
- Type **show ip interface brief** to view interface statuses.

```
C:\>ping 172.16.0.2

Pinging 172.16.0.2 with 32 bytes of data:
Reply from 172.16.0.2: bytes=32 time=54ms TTL=252
Reply from 172.16.0.2: bytes=32 time=44ms TTL=252
Reply from 172.16.0.2: bytes=32 time=55ms TTL=252
Reply from 172.16.0.2: bytes=32 time=43ms TTL=252

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 43ms, Maximum = 55ms, Average = 49ms

C:\>
```

Part 5: Examine the Physical Topology

```
Router0>show ip interface brief
Interface    IP-Address      OK? Method Status  Protocol
FastEthernet0/0  192.168.2.1    YES manual up        up
FastEthernet0/1  10.0.0.1       YES manual up        up
Serial0/0/0     172.31.0.1    YES manual up        up
Serial0/0/1     unassigned     YES unset  administratively down down
Vlan1          unassigned     YES unset  administratively down down
Router0>
```

Step 1: Examine the Cloud.

- Click the **Physical Workspace** tab or press **Shift+P** and **Shift+L** to toggle between the logical and physical workspaces.
- Click the **Home City** icon.
- Click the **Cloud** icon.

How many wires are connected to the switch in the blue rack?

There are 2 wires connected to the switch

- Click **Back** to return to **Home City**.

Step 2: Examine the Primary Network.

- Click the **Primary Network** icon. Hold the mouse pointer over the various cables.

What is located on the table to the right of the blue rack?

A configuration terminal.

- Click **Back** to return to **Home City**.

Step 3: Examine the Secondary Network.

- Click the **Secondary Network** icon. Hold the mouse pointer over the various cables.

Why are there two orange cables connected to each device?

One cable is for receiving data, and the other is for sending/transmitting data (until recently, light could only go one direction on fiber-optic cables, hence why you would need two.)

- Click **Back** to return to **Home City**.

Step 4: Examine the Home Network.

- a. Click the **Home Network** icon.

Why is there no rack to hold the equipment?

They don't have equipment that requires a rack, for example they are using a wireless router.

- b. Click the **Logical Workspace** tab to return to the logical topology.